PADHAKU
India and the Contemporary World – II · Class 10 Social Science



CHAPTER 4

The Age of Industrialisation

NCERT Exercise Solutions

Activity · Introduction	1 Q
Activity · 1.1 The Coming Up of the Factory	1 Q
Activity · 1.2 The Pace of Industrial Change	1 Q
Activity · 2 Hand Labour and Steam Power	1 Q
Discuss · 2.1 Life of the Workers	1 Q
Activity · 3.1 The Age of Indian Textiles	1 Q
Conclusion	1 Q
Write in brief	9 Q
Discuss	4 Q
Project work	1 Q

✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

Independent study aid by Beginers Point. Not an official NCERT publication.

Activity · Introduction

Give two examples where modern development that is associated with progress has led to problems. You may think of environmental issues, nuclear weapons or disease.

SAMPLE ANSWER

- **Environment:** Factories, vehicles and power plants brought progress, but they also caused air and water pollution, global warming and climate change.
- **Nuclear weapons:** Nuclear science gave us electricity, but it also made atom bombs. The bombing of Hiroshima and Nagasaki (1945) killed lakhs of people; accidents like Chernobyl (1986) spread radiation.
- **Disease:** Fast-growing, crowded industrial cities spread diseases; overuse of antibiotics has created drug-resistant germs.

So progress can bring new problems if it is not planned with care.

Activity · 1.1 The Coming Up of the Factory

Historians focus on industrialisation rather than on small workshops: what we believe about the past depends on what historians notice and ignore. Note down one event or

aspect of your own life which adults such as your parents or teachers may think is unimportant, but which you believe to be important.

💡 SAMPLE ANSWER

Model answer: Playing cricket with friends in the evening. Adults may think it is a waste of study time, but for me it is important: it keeps me fit, removes stress, and teaches me teamwork and how to handle winning and losing.

(Write about your own life: a hobby, a friendship, a game, a festival, etc.)

🧪 Activity · 1.2 The Pace of Industrial Change

? **Look at Figs. 4 and 5. Can you see any difference in the way the two images show industrialisation? Explain your view briefly.**

✔ ANSWER

- **Fig. 4 (Lancashire cotton mill, 1925)** shows the mill as grand and beautiful: a huge mill glowing with electricity in the twilight. The artist calls it 'a most impressive sight'. It **glorifies** industrialisation.
- **Fig. 5 (Industrial Manchester, 1857)** shows many chimneys billowing smoke over the city. It shows the **darker side**: smoke, pollution and a crowded, grim landscape.

So Fig. 4 celebrates industry as progress, while Fig. 5 shows its harmful effects on the environment.

🧪 Activity · 2 Hand Labour and Steam Power

? **Imagine that you are a merchant writing back to a salesman who has been trying to persuade you to buy a new machine. Explain in your letter what you have heard and why you do not wish to invest in the new technology.**

💡 SAMPLE ANSWER

Dear Sir,

Thank you for your letter about the new machine. I am sorry, but I do not wish to buy it, for these reasons:

- The machine is very **costly**, and I have heard that it often **breaks down** and repairs are expensive.
- Others say it does not work as well as its makers claim.
- Labour is **plentiful and cheap** here; poor people come from the villages looking for work, so I do not need to save on workers.
- My business is **seasonal**; in the busy months I can hire extra hands and send them away later. A machine would lie idle for part of the year.

- My customers want goods of **many designs and shapes**, which need human skill; the upper classes also prefer handmade goods.

So I will continue with hand labour for now.

Yours faithfully, A Merchant

💬 Discuss · 2.1 Life of the Workers

? Look at Figs. 3, 7 and 11, then reread Source B. Explain why many workers were opposed to the use of the Spinning Jenny.

✔ ANSWER

- In Fig. 3, the whole family spins and **one wheel moves only one spindle**, so many hands were needed.
- With the Spinning Jenny (Fig. 11) and steam-driven wheels (Fig. 7), **one wheel could move many spindles**; one worker could spin several threads at once.
- So the machine **reduced the demand for labour**. Women who lived by hand spinning lost their work.
- Workers already lived in **fear of unemployment**, as work was seasonal and jobs were scarce.
- Source B shows colliers' wives, who had lost work to spinning engines, coming to cut the machine to pieces, believing it would lessen the demand for manual labour.

So workers saw the Jenny as a threat to their livelihood and attacked it.

🧪 Activity · 3.1 The Age of Indian Textiles

? On a map of Asia, find and draw the sea and land links of the textile trade from India to Central Asia, West Asia and Southeast Asia.

💡 SAMPLE ANSWER

Mark these on an outline map of Asia:

- **Land route:** from Punjab, through the north-west frontier and mountain passes, to Afghanistan, eastern Persia (Iran) and Central Asia (goods carried on camel back by Armenian and Persian merchants).
- **Sea route to West Asia:** from **Surat** (Gujarat coast) across the Arabian Sea to the ports of the **Persian Gulf** and the **Red Sea**.
- **Sea route to Southeast Asia:** from **Masulipatam** (Coromandel coast) and **Hoogly** (Bengal) across the Bay of Bengal to Southeast Asian ports.

Draw land routes with one kind of line and sea routes with arrows, and add a key.

⭐ Conclusion

? Look again at Figs. 1 and 2. What would you now say of the images they project?

💡 SAMPLE ANSWER

- Figs. 1 and 2 show a **one-sided, triumphant picture**: industrialisation as only rapid technological progress, and the West as modern while the East is shown as backward.
- The chapter shows that change was **slow and uneven**: new technology was costly and slowly adopted; in Britain even at the end of the 19th century less than 20 per cent of workers were in advanced industries.
- **Hand labour and small-scale production** remained very important, both in Britain and India.
- The images hide the **hardships of workers**: low wages, unemployment, seasonal work and poor living conditions, and the ruin of Indian weavers under colonial rule.

So these images glorify machines but do not tell the full story of industrialisation.

⭐ Write in brief

1 (a) **Explain: Women workers in Britain attacked the Spinning Jenny.**

✔ ANSWER

- The Spinning Jenny (James Hargreaves, 1764) let one worker turn a single wheel and spin several threads at once.
- It speeded up spinning and **reduced the demand for labour**.
- Women who survived on hand spinning feared **losing their jobs**; unemployment was already a big fear.
- So they began attacking the new machines in the woollen industry, and this conflict continued for a long time.

1 (b) **Explain: In the seventeenth century merchants from towns in Europe began employing peasants and artisans within the villages.**

✔ ANSWER

- World trade and colonies increased the **demand for goods**.

- Merchants could not expand production in towns because **urban crafts and trade guilds** were powerful: they controlled production, competition, prices and entry into trade, and rulers gave them monopoly rights.
- In the countryside, open fields were disappearing and commons were being **enclosed**, so poor peasants and cottagers needed new sources of income.
- They eagerly took merchants' advances; they could stay in the village, cultivate their small plots and use all their family labour.

1 (c) **Explain: The port of Surat declined by the end of the eighteenth century.**

✔ ANSWER

- By the 1750s the trade network controlled by Indian merchants was breaking down.
- **European companies** gained power: first concessions from local courts, then **monopoly rights** to trade.
- Exports from Surat fell dramatically, the credit that financed trade dried up, and local bankers went bankrupt. Trade through Surat fell from Rs 16 million (late 17th century) to Rs 3 million by the 1740s.

- New ports, **Bombay and Calcutta**, controlled by European companies and using European ships, grew instead.

1 (d) **Explain: The East India Company appointed gomasthas to supervise weavers in India.**

✔ ANSWER

- After gaining political power, the Company wanted to **eliminate competition**, control costs and ensure regular supplies of cotton and silk goods.
- It removed the existing traders and brokers and wanted **direct control over the weaver**.
- The **gomastha** was a paid servant who supervised weavers, collected supplies and examined the quality of cloth.
- He also made sure that weavers who took Company loans gave their cloth only to the Company.

2 (a) **True or False: At the end of the nineteenth century, 80 per cent of the total workforce in Europe was employed in the technologically advanced industrial sector.**

✔ ANSWER

False. Less than 20 per cent of the total workforce (in Britain) was employed in technologically advanced industrial sectors.

2 (b) **True or False: The international market for fine textiles was dominated by India till the eighteenth century.**

✔ ANSWER

True. Before the age of machine industries, Indian silk and cotton goods dominated the international market; the finer varieties came from India.

2 (c) **True or False: The American Civil War resulted in the reduction of cotton exports from India.**

✔ ANSWER

False. When the American Civil War cut off US cotton supplies, Britain turned to India, and raw cotton exports from India **increased**.

2 (d) **True or False: The introduction of the fly shuttle enabled handloom workers to improve their productivity.**

✔ ANSWER

True. The fly shuttle increased productivity per worker, speeded up production and reduced labour demand.

3 **Explain what is meant by proto-industrialisation.**

✔ ANSWER

Proto-industrialisation is the early phase of industrialisation, before factories, when there was large-scale industrial production for an international market.

- In the 17th and 18th centuries, town merchants gave money (advances) to peasants and artisans in the countryside to produce goods.
- Production was done in **family farms and homes**, not in factories.
- It was controlled by merchants and was part of a network of commercial exchanges; e.g. a merchant clothier bought wool from a stapler and took it to spinners, weavers, fullers and dyers; finishing was done in London.
- It gave poor peasants extra income and fuller use of family labour.

💬 Discuss

1 **Why did some industrialists in nineteenth-century Europe prefer hand labour over machines?**

✔ ANSWER

In Victorian Britain many industrialists preferred hand labour for these reasons:

1. **Plenty of cheap labour:** poor peasants and vagrants came to the cities looking for work, so wages were low and there was no labour shortage.

2. **Costly machines:** machines needed large capital investment, often broke down, repairs were costly, and they were not as effective as claimed.

3. **Seasonal demand:** in gas works, breweries, book-binding, printing and ship repair, work rose and fell with the season, so industrialists hired workers only for the season.

4. **Variety of designs:** the market wanted intricate designs and specific shapes (e.g. 500 kinds of hammers, 45 kinds of axes), which needed human skill; machines made only uniform, standardised goods.

5. **Upper-class taste:** aristocrats and the bourgeoisie preferred handmade goods as a sign of refinement and class; machine-made goods were exported to the colonies.

So, unlike labour-short America, Britain had little reason to replace human hands with machines.

2 **How did the East India Company procure regular supplies of cotton and silk textiles from Indian weavers?**

✔ ANSWER

After establishing political power in Bengal and Carnatic (1760s–70s), the Company built a system of control:

1. **Monopoly:** it asserted a monopoly right to trade, ending competition from French, Dutch, Portuguese and local traders.

2. **Removing middlemen:** it tried to eliminate the existing traders and brokers and control weavers directly.

3. **Gomasthas:** it appointed paid servants called gomasthas to supervise weavers, collect supplies and check the quality of cloth.

4. **System of advances:** weavers were given loans to buy raw material once an order was placed; those who took loans had to hand over their cloth only to the gomastha, not to any other trader.

5. **Full-time weaving:** weavers leased out their land and the whole family worked at weaving.

6. **Force:** gomasthas came with sepoys and peons and punished weavers for delays, often beating and flogging them.

Thus the Company got regular supplies at low prices, but weavers lost their freedom to bargain.

3 **Imagine that you have been asked to write an article for an encyclopaedia on Britain and the history of cotton. Write your piece using information from the entire chapter.**

💡 SAMPLE ANSWER

Britain and the History of Cotton

- **Proto-industrial stage:** In the 17th–18th centuries, merchants gave advances to peasants in the countryside; cloth was made in homes, and finishing was done in London.
- **Inventions and factories:** 18th-century inventions improved carding, twisting, spinning and rolling. Richard Arkwright created the cotton mill, bringing all processes under one roof. The Spinning Jenny (1764) let one worker spin many threads. Raw cotton imports rose from 2.5 million pounds (1760) to 22 million pounds (1787). Cotton was the leading sector up to the 1840s.
- **Workers:** Labour was plentiful and cheap, so hand labour stayed common. Women spinners, fearing unemployment, attacked the Spinning Jenny.

- **Manchester and India:** Britain put import duties on Indian textiles and sold Manchester goods in India. Indian piece-goods fell from 33 per cent of India's exports (1811-12) to 3 per cent (1850-51); cotton piece-goods were over 31 per cent of India's imports by 1850 and over 50 per cent by the 1870s. Indian weavers were ruined. During the American Civil War, Britain bought raw cotton from India.
- **Decline:** During the First World War, British mills made war goods and Manchester imports into India fell. After the war Britain could not compete with the US, Germany and Japan; its cotton production collapsed and cotton exports fell dramatically, while Indian mills captured the home market.

Cotton thus rose and fell with Britain's industrial and colonial power.

4 **Why did industrial production in India increase during the First World War?**

✔ ANSWER

1. British mills were busy with **war production** for the army, so **Manchester imports into India declined**.

2. Indian mills suddenly had a **vast home market** to supply.

3. Indian factories were called upon to supply **war needs**: jute bags, cloth for army uniforms, tents, leather boots, horse and mule saddles and many other items.

4. **New factories** were set up and old ones ran **multiple shifts**.

5. Many new workers were employed and everyone worked longer hours.

So industrial production boomed in the war years, and after the war Manchester never regained its old position in India.

🛠️ Project work

? **Select any one industry in your region and find out its history. How has the technology changed? Where do the workers come from? How are the products advertised and marketed? Talk to employers and workers about the industry's history.**

💡 SAMPLE ANSWER

Model plan:

1. **Choose** an industry near you, e.g. a handloom cluster, sugar mill, brick kiln or garment unit.

2. **History:** when and by whom it was started; how it grew.

3. **Technology:** compare old tools/machines with present ones (e.g. hand loom → power loom → computerised loom).

4. **Workers:** where they come from (nearby villages or other states), working hours, wages, seasonal work.

5. **Marketing:** labels, brand names, newspaper/TV/online ads, shops, exhibitions.

6. **Interviews:** prepare 5–6 questions for one employer and two or three workers; note their views.

7. **Present** your findings with photos, a timeline and a short conclusion.

🎉 **Chapter 4 done! Revise the answers once more tomorrow.**